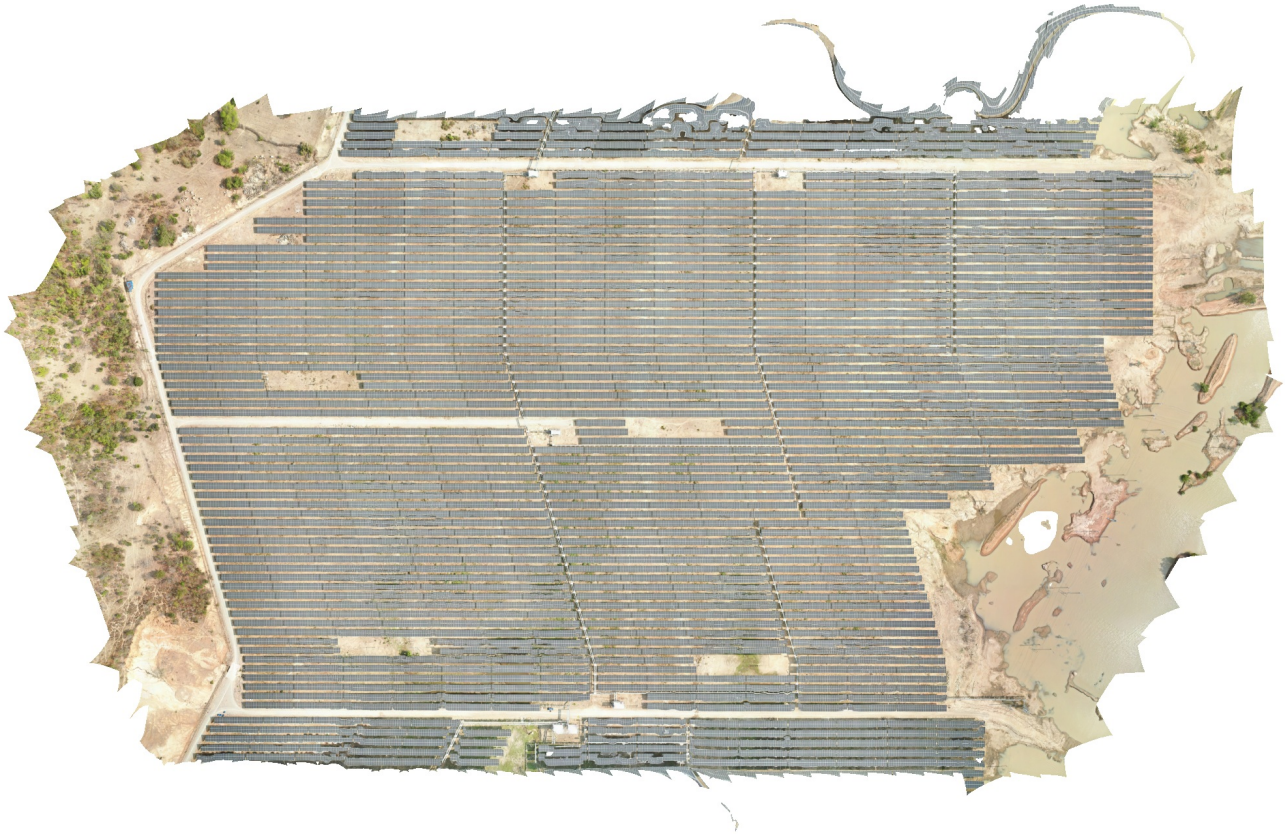
Agisoft Metashape

Processing Report

08 September 2026



Survey Data

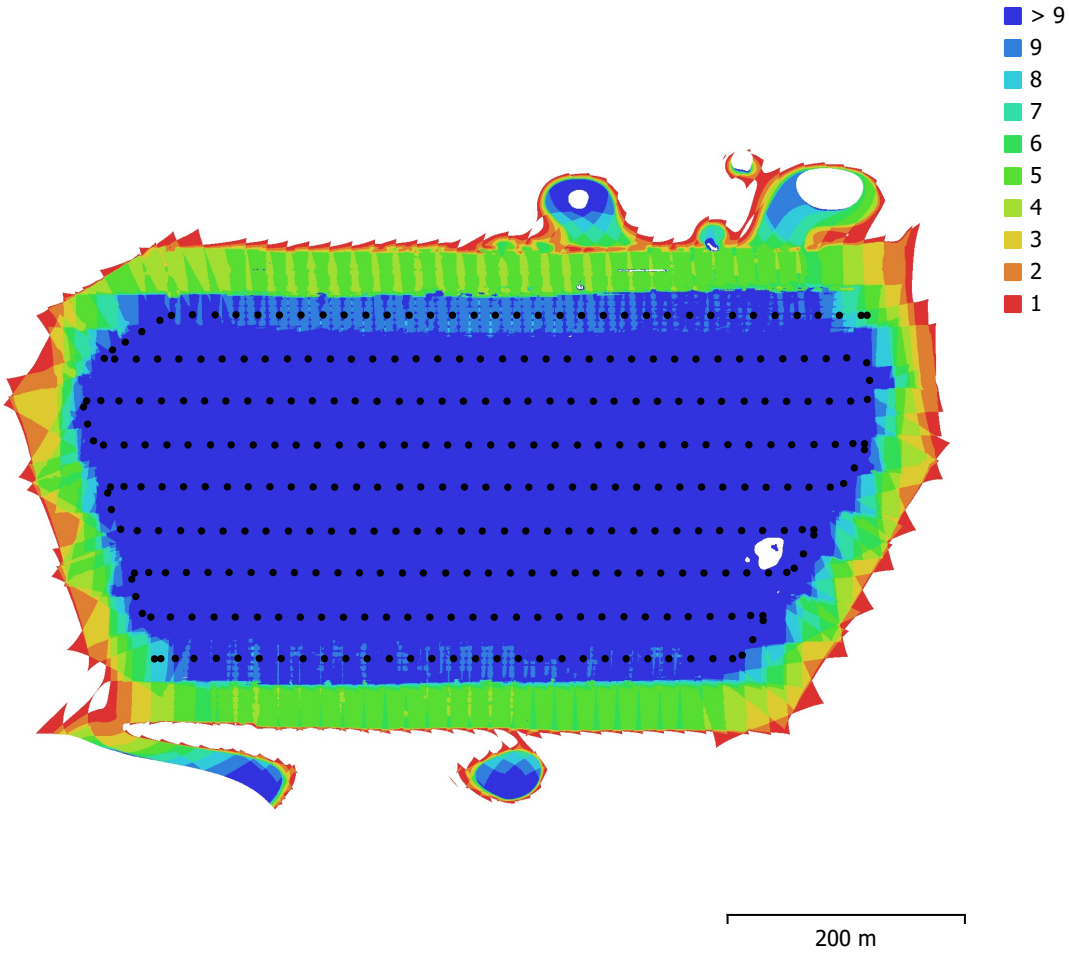


Fig. 1. Camera locations and image overlap.

Number of images:	328	Camera stations:	328
Flying altitude:	84.6 m	Tie points:	228,914
Ground resolution:	2.86 cm/pix	Projections:	672,246
Coverage area:	0.306 km²	Reprojection error:	0.623 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
ZH20T (4.5mm)	4056 x 3040	4.5 mm	1.6 x 1.6 µm	No

Table 1. Cameras.

Camera Calibration

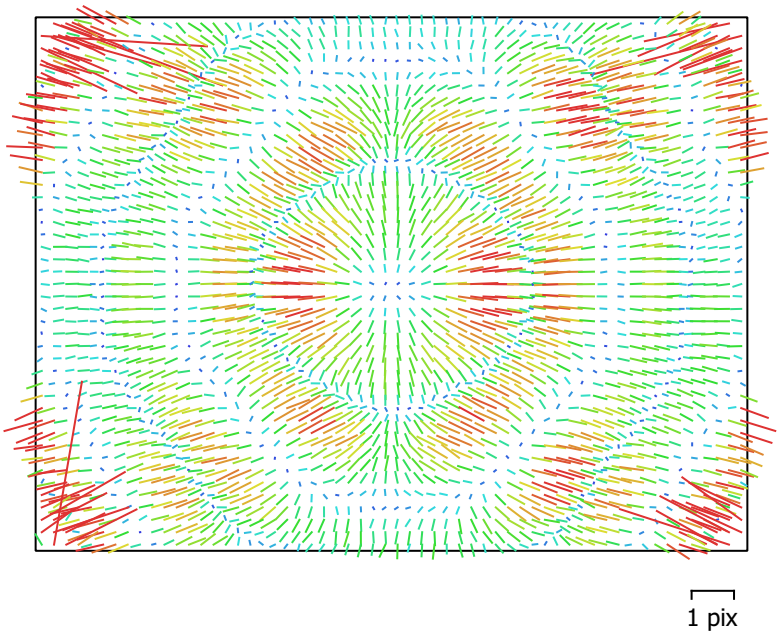


Fig. 2. Image residuals for ZH20T (4.5mm).

ZH20T (4.5mm)

328 images

Type	Resolution	Focal Length	Pixel Size
Frame	**4056 x 3040**	**4.5 mm**	**1.6 x 1.6 µm**

	Value	Error	Cx	Cy	B1	B2	K1	K2	K3	K4	P1	P2
F	**2811.66**											
Cx	**6.40505**	0.08	1.00	0.01	-0.05	-0.07	-0.00	0.03	-0.03	0.03	-0.57	0.01
Cy	**12.3512**	0.065		1.00	0.04	-0.03	-0.01	0.01	-0.00	0.00	0.01	-0.24
B1	**-4.99066**	0.016			1.00	0.05	0.03	-0.03	0.02	-0.02	0.04	-0.02
B2	**-0.333194**	0.016				1.00	0.03	-0.01	0.01	-0.01	0.05	0.04
K1	**0.264551**	0.00016					1.00	-0.84	0.79	-0.78	0.00	0.14
K2	**-0.897653**	0.00064						1.00	-0.99	0.96	0.01	-0.03
K3	**1.14968**	0.0013							1.00	-0.99	-0.01	0.03
K4	**-0.591211**	0.00089								1.00	0.01	-0.04
P1	**0.000172084**	4e-06									1.00	-0.00
P2	**-0.000863895**	5e-06										1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

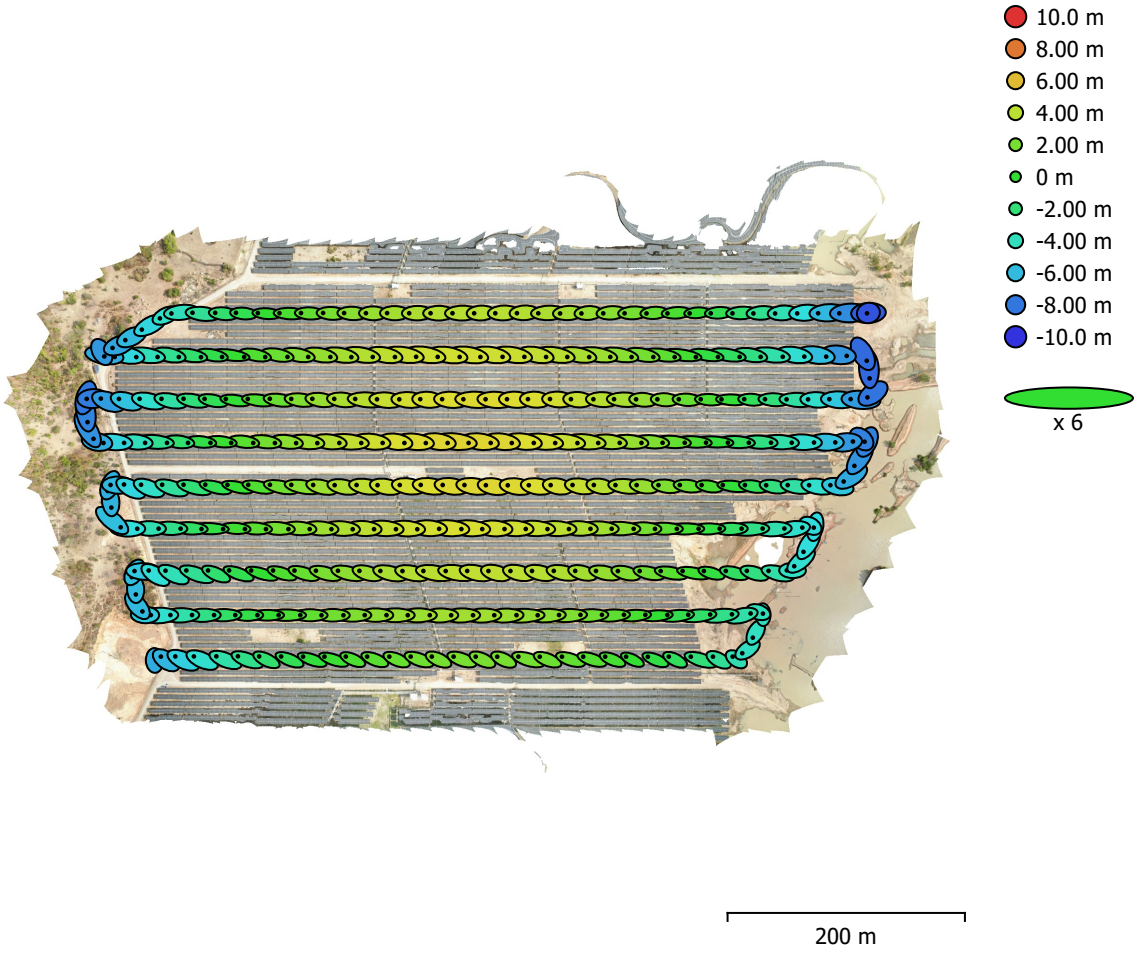


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
2.7228	0.665957	3.85777	2.80306	4.76859

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

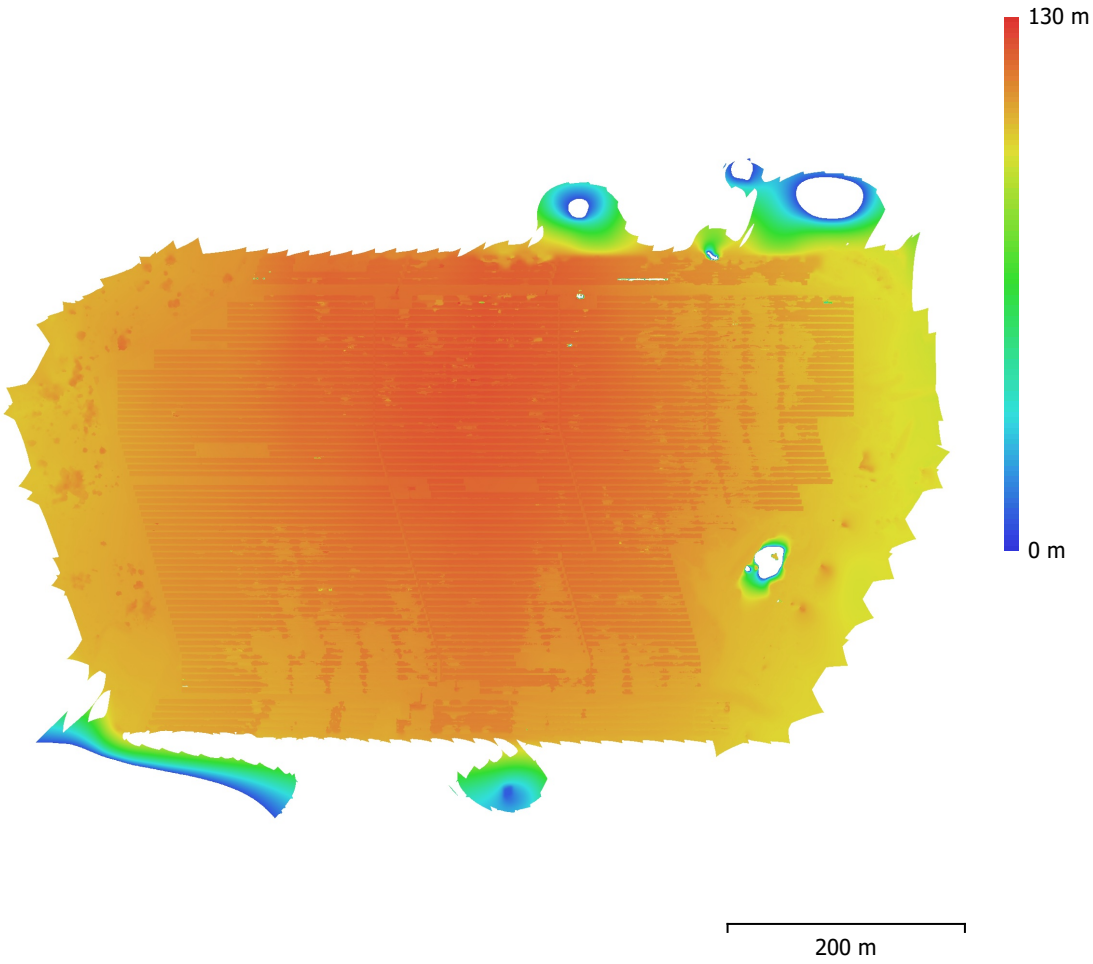


Fig. 4. Reconstructed digital elevation model.

Resolution:	6.02 cm/pix
Point density:	276 points/m²

Processing Parameters

General	
Images	328
Aligned images	328
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	228,914 of 442,935
RMS reprojection error	0.218963 (0.622819 pix)
Max reprojection error	1.36122 (7.46015 pix)
Mean key point size	3.10357 pix
Point colors	3 bands, uint8
Key points	1.05 GB
Average tie point multiplicity	3.18209
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	11 minutes 3 seconds
Matching memory usage	383.74 MB
Alignment time	5 minutes 3 seconds
Alignment memory usage	329.03 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	6 seconds
Date created	2026:09:07 22:26:07
Software version	2.3.0.21954
File size	1.09 GB
Depth Maps	
Count	328
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 31 seconds
Memory usage	2.53 GB
Date created	2026:09:07 22:53:39
Software version	2.3.0.21954
File size	1.19 GB
Point Cloud	
Points	99,501,015
Coordinate precision	1.43 cm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	99,501,015
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 31 seconds
Memory usage	2.53 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	17 minutes 37 seconds
Memory usage	5.62 GB
Date created	2026:09:07 23:11:20
Software version	2.3.0.21954
File size	1.29 GB
DEM	
Size	18,910 x 12,836
Resolution	6.02 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 31 seconds
Memory usage	2.53 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	40 minutes 24 seconds
Memory usage	6.03 GB
Date created	2026:09:07 23:54:09
Software version	2.3.0.21954
File size	832.96 MB
Orthomosaic	
Size	27,648 x 18,944
Resolution	3.01 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	1.73 GB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	9 minutes 52 seconds
Memory usage	3.40 GB
Date created	2026:09:08 00:03:27
Software version	2.3.0.21954
File size	2.23 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU